

Machine Learning–based Late Delivery Risk Prediction in Global Supply Chain Operations

An Exploratory Data-Driven Study of Shipping Performance, Risk Prediction, and Operational Recommendations

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1. Abstract

Late delivery is a major operational risk in global supply chains because a delay at one point in the logistics network can propagate into inventory shortages, service-level agreement (SLA) breaches, customer dissatisfaction, expedited-shipping costs, and inefficient use of warehouse and transportation capacity. This study develops a pre-shipment machine-learning framework for estimating the probability that an order will be delivered late, using the APL Logistics supply-chain dataset.

The source dataset contains **180,519 records and 40 original columns**. After removal of personally identifiable information, high-cardinality fields, redundant fields, and post-shipment variables that would expose the outcome, the modeling dataset is split into **144,415 training observations and 36,104 held-out test observations**. The target, `Late_delivery_risk`, has an overall positive rate of **54.83%**, making late delivery the majority class but not an extreme class-imbalance problem.

The analytical pipeline combines missing-value handling, numerical standardization, categorical one-hot encoding, stratified train-test splitting, and supply-chain-specific feature engineering. Eight engineered variables are used, including a Shipping Pressure Index, Mode Risk Flag, Regional Congestion Index, Order Complexity Score, Order Value per Unit, Profit Margin, High Discount Flag, and Benefit per Unit. A leakage-aware implementation of regional target encoding is included in the supplied `group2_ml.py` pipeline, where the regional encoding is recomputed from the training portion of each cross-validation fold.

Three classifiers are evaluated: Logistic Regression, Random Forest, and XGBoost. On the independent test set, XGBoost produces the strongest discrimination with **ROC-AUC = 0.7745** and an **F1 score = 0.6749** at the standard 0.50 decision threshold. Its binary precision is **84.12%** and recall is **56.35%**. The operational probability framework separates orders into Low (<0.40), Medium ($0.40\text{--}0.70$), and High (≥ 0.70) risk. On the held-out test set, the High-Risk group contains **10,438 orders**, of which **9,361 are actually late**, yielding **89.7% precision**. For the most critical orders (probability ≥ 0.80), precision rises to **95.6%**.

The project also incorporates SHAP explainability and a Streamlit Operations Action Panel, allowing logistics users to inspect risk probabilities, compare regions and shipping modes, and prioritize individual orders. The extended analysis additionally highlights an important interpretation issue: although Standard Class contributes the largest *number* of late orders because of its large volume, the observed late-delivery *rate* is highest for First Class in the supplied dataset. This distinction should be reflected in the final business recommendations.

2. Introduction

2.1 Background

Modern supply chains operate as interconnected networks involving customers, warehouses, transportation services, distribution nodes, and regional markets. Delivery performance therefore depends not only on the physical movement of goods but also on the interaction between

order characteristics, shipping commitments, geography, commercial conditions, and operational capacity.

A conventional monitoring system usually identifies a delivery problem after the event has already occurred. Such a reactive approach limits the available intervention window. Once an order is already in transit and has missed its planned schedule, options such as rerouting, reprioritization, carrier escalation, customer notification, or warehouse intervention may be more expensive and less effective.

The objective of this project is consequently different from retrospective delay analysis. The central question is:

Which orders are likely to experience a late delivery before shipment occurs?

A probability-based answer allows the organization to allocate scarce operational resources according to risk rather than treating every order equally.

2.2 Problem Statement

The project requirements identify three major operational challenges:

1. Unpredictable shipment delays.
2. High operational costs associated with last-minute corrective actions.
3. Limited ability to prioritize orders according to their probability of failure.

The proposed system addresses these challenges by producing:

- a continuous late-delivery probability between 0 and 1;
- a discrete Low/Medium/High risk classification;
- model-based explanations of important risk drivers; and
- an operational queue of high-risk orders.

The supplied project specification explicitly emphasizes prediction **before shipment**, rather than explaining an outcome using information that only becomes available after shipment.

2.3 Research Contributions

The extended study makes the following practical contributions:

- establishes a leakage-controlled pre-shipment prediction boundary;
- combines transactional, geographic, commercial, and shipping variables;
- engineers domain-specific indicators rather than relying only on raw columns;
- compares linear and non-linear machine-learning models;
- translates probabilities into operational risk tiers;
- reports both model-level and business-level precision;

- incorporates SHAP explanations for individual decisions;
- provides a Streamlit-based operational interface; and
- identifies data-quality and interpretation issues that should be addressed before production deployment.

3. Related Work and Methodological Position

The project specification positions the work within supply-chain analytics and predictive logistics. Earlier analytical approaches frequently focus on historical performance, transit duration, or post-shipment status. Those variables can be highly predictive, but they may not be available when an operational decision must be made.

The key methodological distinction in this study is therefore the **pre-shipment information boundary**.

For example:

- `Days for shipping (real)` represents actual shipping duration and

is only known after the event.

- `Delivery Status` describes the realized delivery state.
- `Order Status` may contain information about the eventual state of the order.

Including such fields would allow the model to learn from information that is unavailable at prediction time. The supplied preprocessing pipeline therefore removes them before modeling.

The study also extends predictive modeling with SHAP, which provides a feature-level explanation of model output rather than presenting a probability as an unexplained black-box score.

4. Dataset Description

4.1 Source Dataset

The supplied APL Logistics CSV contains:

- **180,519 records**
- **40 original columns**
- **1 binary target variable**
- a mixture of numerical and categorical variables.

The project guide describes variables covering payment type, scheduled shipping duration, financial measures, customer information, geographic information, product information, shipping mode, and delivery outcomes.

The target variable is:

`Late_delivery_risk`

where:

- **1** = late delivery;
- **0** = not late.

The observed target distribution in the supplied CSV is:

Target Records Share ----- On-Time 81,542
45.17% Late 98,977 54.83% **Total 180,519 100.00%**

Thus, late deliveries form the majority class.

4.2 Modeling Boundary

The modeling dataset intentionally excludes three types of information:

Post-shipment leakage - `Days for shipping (real)` - `Delivery Status` - `Order Status`

Personally identifiable information - customer first and last names; - street address; - customer ZIP code; - customer identifiers.

High-cardinality or deployment-unhelpful location/product fields - `Product Name`; - `Order City`; - `Order State`; - `Order Country`; - `Customer Country`; - `Customer State`.

This creates a cleaner feature space and reduces the risk that the model memorizes individual entities rather than learning generalizable operational patterns.

4.3 Missing Values

The supplied dataset contains missing values in a small number of fields. In particular, the project preprocessing identifies missing values in `Customer Lname` and `Customer Zipcode`.

The modeling pipeline uses:

- median imputation for numerical variables;
- most-frequent imputation for categorical variables.

The use of a unified preprocessing pipeline ensures that the same transformations can be applied during inference.

5. Data Preprocessing

5.1 Stratified Train-Test Split

The dataset is divided using an **80/20 stratified split** with random state 42.

Dataset Records ----- Training 144,415 Test 36,104 Total 180,519

The training set contains 79,181 late and 65,234 on-time records.

The held-out test set contains:

- **19,796 late orders**
- **16,308 on-time orders**

Stratification preserves the approximate target distribution between training and test partitions.

5.2 Numerical Processing

The supplied pipeline uses:

`SimpleImputer(strategy="median") → StandardScaler()`

This is applied to the numerical feature path.

5.3 Categorical Processing

Categorical variables are processed through:

`SimpleImputer(strategy="most_frequent") → OneHotEncoder(handle_unknown="ignore")`

The `handle_unknown="ignore"` configuration is important for deployment because a future order may contain a category that was not present in the training data.

5.4 Pipeline Integrity

The project uses a Scikit-Learn `ColumnTransformer` and complete model pipelines. This prevents preprocessing operations such as scaling and encoding from being performed inconsistently between training and inference.

The supplied project metadata reports **24 numerical features and 7 categorical features** after feature engineering, with **124 transformed features** after encoding.

6. Feature Engineering

The project goes beyond raw variables by creating eight derived indicators.

6.1 Shipping Pressure Index

The supplied implementation defines:

$$\backslash \text{ShippingPressureIndex} = \backslash \text{frac}\{\text{DaysForShipmentScheduled}\}=\text{tex}\{\text{OrderItemQuantity} + 1\} \backslash$$

The purpose is to represent the scheduled time available relative to order quantity. The +1 denominator offset avoids division by zero and moderates the scale for small quantities.

6.2 Mode Risk Flag

The supplied implementation groups:

- First Class and Same Day → 0
- Second Class and Standard Class → 1

This feature provides a coarse distinction between express-like and economy-like modes. However, the raw shipping-mode one-hot variables are also retained, allowing the XGBoost model to learn mode-specific differences.

6.3 Regional Congestion Index

The Regional Congestion Index is a target-encoded estimate of historical late-delivery frequency for each Order Region.

For a region (r):

$$[\text{RegionalCongestionIndex}(r) = \frac{\text{Late orders in } r}{\text{All orders in } r}]$$

The mapping is calculated from training data and a global mean is used as the fallback for unseen regions.

The supplied training run maps **23 regions**, with a global fallback mean of approximately **0.5483**.

6.4 Order Complexity Score

The implementation combines normalized quantity, discount rate, and scheduled days:

$$[\text{Complexity} = \frac{\text{Quantity}}{5} + \text{DiscountRate} + \frac{\text{ScheduledDays}}{4}]$$

This produces a composite indicator of order characteristics that may increase operational complexity.

6.5 Additional Financial and Commercial Features

Four additional variables are created:

Order Value per Unit

$$[\text{OrderValuePerUnit} = \frac{\text{Sales}}{\text{OrderItemQuantity}+1}]$$

Profit Margin

$$[\text{ProfitMargin} = \frac{\text{OrderProfitPerOrder}}{\text{Sales}+1}]$$

High Discount Flag

$$[\text{HighDiscountFlag} = I(\text{DiscountRate} > 0.10)]$$

Benefit per Unit

$$[\text{BenefitPerUnit} = \frac{\text{BenefitPerOrder}}{\text{OrderItemQuantity}+1}]$$

These features provide the model with relative measures rather than only absolute monetary values.

7. Exploratory Data Analysis

7.1 Overall Late-Delivery Risk

The overall late-delivery rate is **54.83%**, indicating that the historical process contains a substantial delivery-risk problem.

This also establishes an important business baseline: a system that simply predicts every order as late would have high recall but poor operational usefulness. The objective is instead to distinguish higher-risk orders from lower-risk orders.

7.2 Shipping Mode Analysis

A detailed calculation from the supplied CSV reveals the following:

Shipping Mode	Orders	Late Rate
Standard Class	107,752	41,023 38.07%
Second Class	35,216	26,987 76.63%
First Class	27,814	26,513 95.32%
Same Day	9,737	4,454 45.74%

This finding requires an important interpretation distinction.

Standard Class accounts for the largest absolute number of late orders because it represents the largest share of all orders. However, First Class has the highest observed late-delivery rate.

The same pattern is visible in scheduled shipping days because First Class corresponds to one scheduled day in the supplied data, whereas Standard Class corresponds to four scheduled days.

Therefore, the research paper should not describe Standard Class as having the highest *late rate*. A more accurate conclusion is that **shipping mode and scheduled duration are dominant structural predictors, while high-volume Standard Class generates the largest absolute number of late cases.**

7.3 Scheduled Shipping Duration

The target rate varies substantially by scheduled duration:

Scheduled Days	Orders	Late Rate
0	9,737	45.74%
1	27,814	95.32%
2	35,216	76.63%
4	107,752	38.07%

This is one of the strongest empirical patterns in the dataset. Very short scheduled windows are associated with substantially higher late-delivery incidence.

The finding supports the use of scheduled duration as a central predictor, but it also indicates that the model is partly learning the operational difficulty embedded in the original scheduling policy.

7.4 Regional Risk

The highest observed historical late rates in the raw dataset include:

Region	Orders	Late Rate
Central Africa	1,677	57.96%
South Asia	7,731	56.27%
East Africa	1,852	55.94%
Western Europe	27,109	55.85%
South of USA	4,045	55.77%

Central Africa is approximately three percentage points above the global baseline, while South Asia is approximately 1.4 percentage points above it.

The model's regional-risk output similarly places Central Africa, Eastern Europe, South Asia, and South of USA among the higher-risk regions in the held-out test predictions.

7.5 Customer Segment

The raw-data late rates are relatively close across customer segments:

Customer Segment	Orders	Late Rate
Home Office	32,226	55.07%
Consumer	93,504	54.81%
Corporate	54,789	54.72%

This suggests that customer segment is unlikely to be as operationally decisive as shipping mode and scheduled duration.

7.6 Discount Rate

The supplied data does not show a simple monotonic relationship between discount rate and delay risk. For example, the observed late rate is approximately 54.87% for orders with a 0--5% discount and approximately 54.57% for orders with a 5--10% discount.

The highest discount band above 20% has a late rate of approximately 55.67%.

Consequently, the relationship between discounts and late delivery should be described as **weak or context-dependent**, rather than as a direct causal effect.

8. Model Development

Three models are evaluated.

8.1 Logistic Regression

Logistic Regression serves as an interpretable linear benchmark. It establishes how well a relatively simple additive model can distinguish late from on-time orders.

8.2 Random Forest

Random Forest is used to capture non-linear interactions and threshold effects that Logistic Regression may miss.

The supplied implementation uses:

- 300 trees;
- maximum depth of 20;
- minimum leaf size of 5;
- square-root feature selection;
- balanced class weights.

8.3 XGBoost

XGBoost is the selected model because it achieves the highest held-out ROC-AUC.

The supplied configuration uses:

- 400 estimators;
- maximum depth of 8;
- learning rate of 0.05;
- 0.8 row subsampling;
- 0.8 column subsampling;
- minimum child weight of 5;
- $\gamma = 0.1$;
- regularization through α and λ ;
- class-weight adjustment through `scale_pos_weight`.

9. Cross-Validation and Leakage Control

The project initially identified a methodological risk in regional target encoding. If the regional late-delivery rate is calculated using all training labels before cross-validation, the validation records indirectly influence their own encoded feature.

The supplied `group2_ml.py` contains a corrected implementation. Its `cv_score()` function:

1. creates each StratifiedKFold split;
2. extracts the fold's training subset;
3. computes the regional target map using only that subset's labels;
4. engineers the regional feature for both fold-training and fold-validation data using that map;
5. clones a fresh model pipeline;
6. fits on the fold-training data; and
7. evaluates AUC on the validation data.

This is the correct direction for leakage control.

However, the stored project artifacts should clearly record whether the reported CV summary was generated after this correction. The **held-out test set remains the primary unbiased reference for the final reported performance**, because it was not used to construct the final regional map.

10. Model Evaluation

10.1 Test-Set Performance

```
Model ROC-AUC Precision Recall F1 -----
----- Logistic Regression 0.7405 0.8499 0.5420
0.6619 Random Forest 0.7543 0.8450 0.5517 0.6676 XGBoost 0.7745
0.8412 0.5635 0.6749
```

All binary precision, recall, and F1 values above use the standard **0.50 threshold**.

XGBoost improves ROC-AUC by:

- 0.0340 over Logistic Regression;
- 0.0202 over Random Forest.

Its F1 score is also the highest.

10.2 Confusion Matrix

For XGBoost at the 0.50 binary threshold:

```
Predicted On-Time Predicted Late -----
----- Actual On-Time 14,202 2,106 Actual Late 8,640 11,156
```

This gives:

- True Negatives = 14,202
- False Positives = 2,106
- False Negatives = 8,640
- True Positives = 11,156

The resulting accuracy is approximately **70.24%**.

Specificity is approximately **87.09%**, while the false-negative rate is approximately **43.65%**. The latter is operationally important: although positive predictions are relatively precise, a substantial number of actual late orders remain below the standard 0.50 threshold.

This is one reason why the continuous probability score and high-risk operational tier are more useful than treating the 0.50 cutoff as the only business decision rule.

10.3 Cross-Validation Results

The stored project results report:

```
Model Mean CV AUC Standard Deviation -----
----- Logistic Regression 0.7397 0.0025 Random Forest
0.7503 0.0017 XGBoost 0.7683 0.0018
```

XGBoost again ranks first.

The supplied source code now implements per-fold target encoding, so the reproducibility report should distinguish the corrected implementation from any earlier artifact generated before the correction.

11. Risk Probability and Operational Classification

A classifier's probability is more useful operationally when converted into decision bands.

The project defines:

- **Low Risk:** probability < 0.40
- **Medium Risk:** $0.40 \leq \text{probability} < 0.70$
- **High Risk:** probability ≥ 0.70

11.1 Test-Set Tier Results

```
Risk Tier Test Orders Actually Late Precision -----
----- Low 18,931 6,719 35.5% Medium 6,735 3,716
55.2% High 10,438 9,361 89.7%
```

The High-Risk group contains **28.9% of the test orders** but captures a very high concentration of actual late cases.

For probabilities ≥ 0.80 :

- 6,574 orders are flagged;
- 6,282 are actually late;
- precision = **95.6%**.

This creates a useful hierarchy of interventions:

Probability Suggested operational response -----
----- \<0.40 Standard monitoring
0.40--0.70 Proactive watchlist 0.70--0.80 Immediate review ≥0.80
Critical intervention / expedited handling

These are operational recommendations based on the supplied project's performance outputs; the exact intervention policy should ultimately be validated against cost, capacity, and SLA requirements.

12. Probability Calibration

The project reports a Brier Score of **0.1885**, compared with an uninformative baseline of **0.2477**.

A lower Brier Score indicates better probabilistic accuracy.

The supplied calibration observations include:

Predicted Probability Band Observed Late Rate -----
----- 0--20% 15.2% 40--60% 51.4% 80--100% 95.6%

The close relationship between predicted risk and observed event frequency supports the use of probability bands for operational prioritization.

Nevertheless, calibration should be monitored after deployment because changes in customer mix, network capacity, shipping policies, seasonality, or external disruptions can cause probability drift.

13. Explainability with SHAP

The project uses SHAP with TreeExplainer for XGBoost.

13.1 Global Importance

The supplied XGBoost feature-importance output identifies the following leading predictors:

Rank Feature Importance ----- 1
Shipping Mode_Standard Class 0.3663 2 Mode Risk Flag 0.1904 3
Shipping Mode_Same Day 0.1484 4 Days for shipment (scheduled)
0.1223 5 Shipping Mode_First Class 0.0610 6 Type_TRANSFER
0.0080 7 Shipping Mode_Second Class 0.0071 8 Shipping Pressure
Index 0.0029 9 Type_CASH 0.0016 10 Type_PAYMENT 0.0015

This confirms that **shipping configuration and scheduled duration dominate the learned model**.

A critical methodological point is that feature importance indicates predictive association, not causation. For example, a high importance for Standard Class does not by itself prove that Standard Class causes late delivery.

13.2 Local Explanations

The supplied SHAP evaluation demonstrates order-level explanations in which individual feature contributions are expressed as positive or negative impacts on the model output.

This enables the dashboard to answer a practical question:

Why was this particular order flagged as high risk?

Instead of displaying only "High Risk," the interface can identify the variables most responsible for increasing or decreasing the score.

14. Business Interpretation of Shipping-Mode Findings

The raw dataset and model outputs together require careful interpretation.

The dataset shows:

- First Class: 95.32% late;
- Second Class: 76.63% late;
- Same Day: 45.74% late;
- Standard Class: 38.07% late.

Therefore, the statement that Standard Class has the highest *rate* of late delivery is not supported by the supplied CSV.

However, Standard Class represents approximately **59.7% of all records**, so it still contributes the largest absolute number of late orders.

A stronger business interpretation is:

Shipping mode is a dominant risk discriminator, but the relationship is not equivalent to "slower service always means higher observed late rate." Short scheduled windows appear particularly difficult to satisfy in this dataset.

This finding is operationally more useful because it points toward **schedule design and SLA feasibility**, not merely carrier speed.

15. Regional Risk and Capacity Planning

Regional aggregation allows the individual probabilities to be converted into network-level planning information.

The supplied test predictions generate regional measures including:

- total orders;
- mean predicted late probability;
- high-risk order count;
- percentage of orders classified as high risk.

For example, the generated regional output places Central Africa at approximately **0.552 mean predicted probability**, with **33.1%** of its test orders classified as High Risk.

Other high-risk regional groups include Eastern Europe, South Asia, South of USA, Southeast Asia, and US Center.

Regional risk should not automatically be interpreted as infrastructure failure. Differences can reflect:

- shipping-mode mix;
- scheduled-duration policies;
- order composition;
- customer/product mix;
- historical operational conditions.

Therefore, regional risk maps should be used to prioritize investigation and resource planning rather than to assign blame to a geographic node.

16. Streamlit Operations Action Panel

The supplied application operationalizes the model through Streamlit.

16.1 Delay Risk Overview

The dashboard provides macro-level indicators such as:

- total orders;
- risk distribution;
- high-risk order count;
- overall risk profile.

16.2 Order-Level Prediction

Users can enter or inspect an individual order and receive:

- predicted late probability;
- Low/Medium/High risk category;
- contributing factors through SHAP.

16.3 Region and Shipping-Mode Analysis

The dashboard supports:

- regional risk visualization;
- shipping-mode comparison;

- filtering by market and customer segment.

16.4 Operations Action Queue

The Action Panel converts the model into a workflow.

A practical priority ordering is:

1. probability ≥ 0.80 ;
2. probability $0.70-0.80$;
3. probability $0.40-0.70$;
4. probability < 0.40 .

This supports differentiated interventions rather than applying the same treatment to every order.

17. Proposed Operational Workflow

A production implementation can follow this sequence:

Order placement

↓

Pre-shipment data capture

↓

Data validation and preprocessing

↓

Feature engineering

↓

XGBoost probability prediction

↓

Risk classification

↓

SHAP explanation

↓

Operational action

↓

Outcome capture

↓

Model monitoring and retraining

This closes the loop between prediction and operational learning.

For example:

- High-risk orders can receive manual review.
- Critical orders can be considered for premium shipping or reprioritization.
- Regional concentrations can trigger staffing or capacity planning.
- Medium-risk orders can receive proactive monitoring.
- Low-risk orders can follow normal workflows.

18. Business Recommendations

Recommendation 1: Use Probability-Based Prioritization

Do not treat the 0.50 binary threshold as the only decision rule. The ≥ 0.70 operational tier has substantially higher precision and is more aligned with intervention decisions.

Recommendation 2: Establish a Critical ≥ 0.80 Queue

The ≥ 0.80 group has **95.6% precision** in the supplied test set. This makes it a strong candidate for scarce, high-cost interventions, subject to business validation.

Recommendation 3: Audit Scheduling Policies

The strong relationship between scheduled duration and observed late-delivery rate suggests that SLA and scheduling policy deserve direct investigation.

In particular, orders with one- or two-day schedules exhibit much higher historical late rates than four-day schedules.

Recommendation 4: Do Not Assume Shipping-Class Labels Reflect Actual Risk

The raw data demonstrates that First Class has the highest observed late rate. Operational teams should therefore analyze the relationship among shipping mode, scheduled days, and actual network performance instead of assuming that a service label automatically represents low risk.

Recommendation 5: Use Regional Predictions for Capacity Planning

Regional heatmaps should guide:

- warehouse staffing;
- dispatch prioritization;
- exception management;
- capacity allocation;
- regional performance reviews.

Recommendation 6: Integrate with WMS/OMS

The prediction pipeline can be integrated with warehouse or order-management systems so that a risk score is generated automatically at order placement.

Recommendation 7: Monitor Model Drift

Production monitoring should track:

- ROC-AUC;
- precision and recall;
- high-risk precision;
- calibration/Brier score;
- feature distribution drift;
- regional performance drift;
- shipping-mode performance drift.

Recommendation 8: Preserve the Leakage Boundary

Any future data source should be classified as either:

1. available at order placement;
2. available before dispatch; or
3. available only after shipment.

Only the first two categories should be included if the objective remains pre-shipment prediction.

19. Limitations

19.1 External Dynamic Variables

The model does not include live external variables such as:

- weather;
- port congestion;
- strikes;

- traffic;
- geopolitical disruption;
- real-time transportation capacity.

Such variables could improve responsiveness to sudden disruptions.

19.2 Carrier Identification

The dataset does not provide explicit carrier identification. Consequently, the model cannot directly determine which logistics provider caused a delay.

19.3 Causal Interpretation

The model identifies predictive relationships, not causal effects. A feature with high SHAP importance should not automatically be interpreted as the root cause of delay.

19.4 Short Scheduled Windows

Because scheduled shipping duration is highly predictive, the model may partly encode the organization's existing scheduling policy. This is useful for prediction but should be considered when interpreting the model as an explanation of underlying logistics causes.

19.5 Cross-Validation Artifact Management

The supplied source code contains a corrected fold-specific target-encoding procedure. Research reporting should ensure that all stored CV metrics are explicitly tied to the exact code version that generated them.

19.6 Temporal Validation

The current evaluation uses a random stratified train-test split. For a genuinely forward-looking deployment, an additional **time-based validation experiment** would be valuable, because a random split may mix patterns from different periods.

20. Future Research

Several extensions could substantially strengthen the study.

20.1 Temporal Validation

Sort records chronologically and train on earlier periods while testing on later periods. This would more closely simulate real deployment.

20.2 Cost-Sensitive Threshold Optimization

The 0.70 and 0.80 thresholds are useful operational bands, but an optimal threshold should be determined from:

- cost of a missed late order;
- cost of an unnecessary intervention;
- cost of premium shipping;
- customer/SLA penalties.

A formal expected-cost framework could identify the economically optimal intervention threshold.

20.3 Advanced Calibration

Platt scaling or isotonic calibration could be tested if future validation demonstrates systematic probability miscalibration.

20.4 External Data Integration

Future versions could integrate weather, port, traffic, holiday, capacity, and geopolitical indicators.

20.5 Fairness and Segment Monitoring

Performance should be evaluated separately across markets, customer segments, and regions to ensure that a strong aggregate AUC does not conceal weak performance in particular operational groups.

20.6 Model Monitoring

A production model should trigger retraining when:

- feature distributions change;
- calibration deteriorates;
- high-risk precision falls;
- new shipping policies are introduced;
- regional operational conditions change.

21. Conclusion

This study demonstrates the feasibility of transforming late-delivery management from a reactive process into a predictive risk-intelligence workflow.

Using **180,519 supply-chain records**, the project constructs a leakage-controlled modeling dataset and compares Logistic Regression, Random Forest, and XGBoost. XGBoost provides the strongest held-out discrimination with **ROC-AUC = 0.7745**, precision of **84.12%**, recall of **56.35%**, and F1 score of **0.6749** at the standard 0.50 threshold.

The operational probability framework provides an additional business perspective. At the ≥ 0.70 threshold, **10,438 of 36,104 test orders** are classified as High Risk, with **9,361 actually late**, yielding **89.7% precision**. At ≥ 0.80 , precision reaches **95.6%**.

The analysis also shows that shipping mode and scheduled duration are the dominant predictive factors. Importantly, the raw data demonstrates that Standard Class contributes the largest number of late cases because of its volume, while First Class has the highest late-delivery rate. This distinction should guide the final business interpretation and prevent an incorrect conclusion that Standard Class has the highest rate of failure.

The Streamlit dashboard, SHAP explanations, regional risk analysis, and action queue provide the operational layer required to turn model probabilities into decisions. The next stage should focus on time-based validation, cost-sensitive threshold selection, external dynamic data, and continuous production monitoring.

Overall, the project establishes a practical predictive backbone for intelligent supply-chain operations while highlighting the importance of strict leakage control, careful statistical interpretation, and alignment between model outputs and real operational costs.

22. References

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26. APL Logistics project specification and dataset documentation supplied with the project files.
27. APL Logistics late-delivery prediction project code, evaluation reports, model metadata, and generated outputs supplied with the project files.